I/P: [1, 1, 2, 3, 3, 3, 4, 7]

O/P: [1, 2, 3, 4, 7]

[1, 1, 2, 3, 3, 3, 4, 7]

[1, 2, 3, 4, 7]

TC: O(n)
AS: O(1)

0 1 2 3 4 5 6 7
[1, 2, 3, 4, 7, x, x, x]

return 5.

responsible for getting the first occ of any element.

[1, 2, ~~1~~, 3, ~~2~~, 4, ~~3~~, 7, ~~3~~, 3, 4, 7]

responsible for telling where the new element will come.

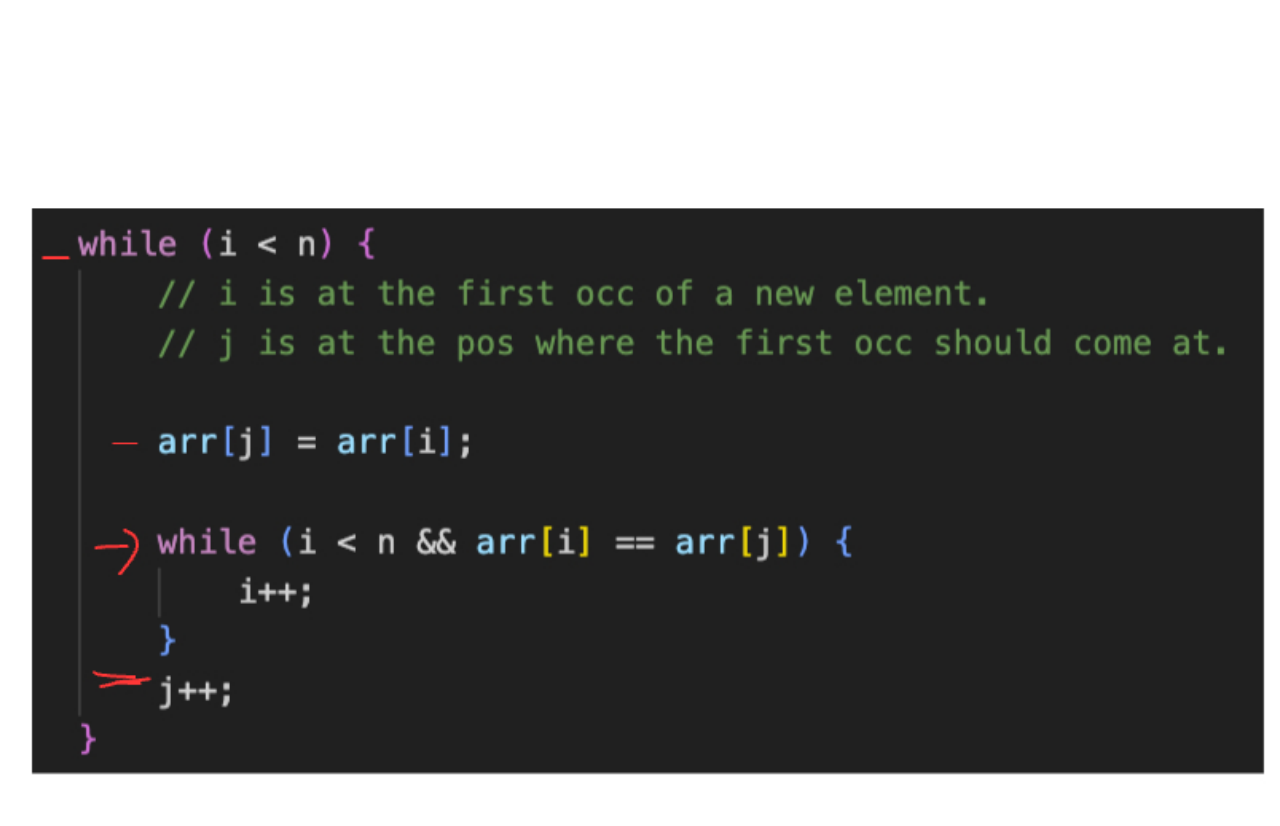
~~arr[i]~~ = arr[j].

arr[j] = arr[i].

TC: ______

AS: O(1).

0 1 2 3 4 5 6 7
[1, 1, 2, 3, 3, 3, 4, 7].



```
while (i < n) {
    // i is at the first occ of a new element.
    // j is at the pos where the first occ should come at.

    arr[j] = arr[i];

    while (i < n && arr[i] == arr[j]) {
        i++;
    }
    j++;
}
```

[1, 2, 3, 4, 7, 3, 4, 7]

i: 0 ⟶ n-1

j: 0 ⟶ n-1

i = 0
j: 0 - - - n-1
i = 1
j: 0 - - - n-1
i = 2
j: 0 - - - n-1

Binary Search

Divide & Conquer.

[1, 3, 5, 8, 11, 15, 25],

target: 11

[11, 15, 25].

[11].

[1, 3, 5, 8, 11, 15, 25]   targ = 10.

[11, 15, 25]

[11].

[ ]

n
↓
n/2
↓
n/4
⋮
↓
1

no. of iterations = O(log n).

$\underbrace{1, 2, 4, 8 \dots\dots, n}_{x}$

a = 1
r = 2

$n = a r^{x-1}$

$n = 1 \cdot 2^{x-1}$

$x - 1 = \log_2 n$

$x = O(\log n)$

0 1 2 3 4 5 6
[1, 3, 5, 8, 11, 15, 25]   targ = 11

low   high

mid = (low + high) / 2

low=0, high=6
mid = 3
Disc. the LH.
low = mid+1

low=4 high=6
mid = (4+6)/2 = 5
Disc. the RH
high = mid-1

low = 4 high = 4
mid = 4

---

0 1 2 3 4 5 6 7 8
[1, 3, 5, 5, 5, 5, 67, 123, 125]
targ = 5

firstOcc = 2
Last Occ = 5

low = 0   high = 8
mid = 4.

fin Occ = ~~4~~ 2

low = 0   high = 3
mid = (0+3)/2 = 1

TC = O(log n)
AS = O(1).

low = 2   high = 3
mid = (2+3)/2 = 2

low = 2   high = 1

N = 30.

i = 1   1
= 2   4
= 3   9
= 4   16
= 5   25
= 6   36 > 30
7
8
⋮

TC = O(sqrt(n))

res = ~~3~~ 5

N = 30

low = 1   high = N = 30
mid = (1+30)/2 = 15.   Disc. RH.

low = 1   high = 14
mid: 15/2 = 7   Disc. RH.

low = 1   high = 6.
mid = 7/2 = 3.   Disc. LH.

low = 4   high = 6.
mid = 10/2 = 5.   Disc. LH.

low = 6   high = 6.
mid = 6.   Disc. RH.

low = 6   high = 5.   STOP!

TC = O(log n).
AS = O(1).

N = 30   O/P = 5.4.

floorValue = 5

5.0, 5.1, 5.2, 5.3 - - - - - - 5.9

low = 0   high = 9
mid = 9/2 = 4.

5 + 4/10 = 5.4

$(fv + \frac{mid}{10})^2$ N.